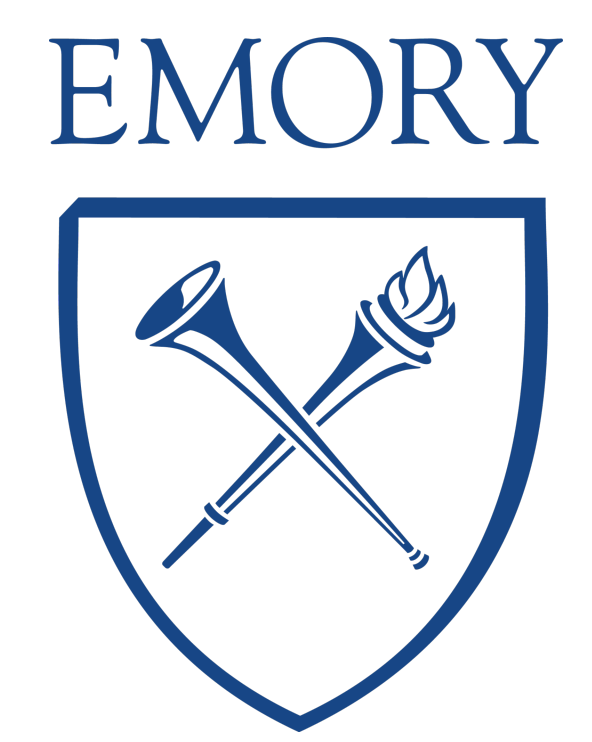




Revisiting Inhibitory Control Measurement: Evaluating Task Selection, Reliability, and Trial Count Effects



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Introduction

Inhibitory Control

Inhibitory control (IC) is a core executive function that focuses on the ability to actively inhibit or delay a dominant response to achieve a goal¹

- Researchers have typically assessed IC using a variety of different laboratory tasks and outcome measures²
- This high variability across studies has led to many inconsistencies in IC research³
- Potential factors contributing to inconsistencies in IC assessment:

1. The inconsistent use of tasks⁴
2. The inconsistent use of outcome measures⁵
3. Low reliability in outcome measures⁶

RQ: What is the best way to measure inhibitory control?

Method

Participants

- 166 individuals ($M_{age} = 19.10$, $SD = 1.03$, range = 18–22) completed 250 trials of each task

Experiment Design

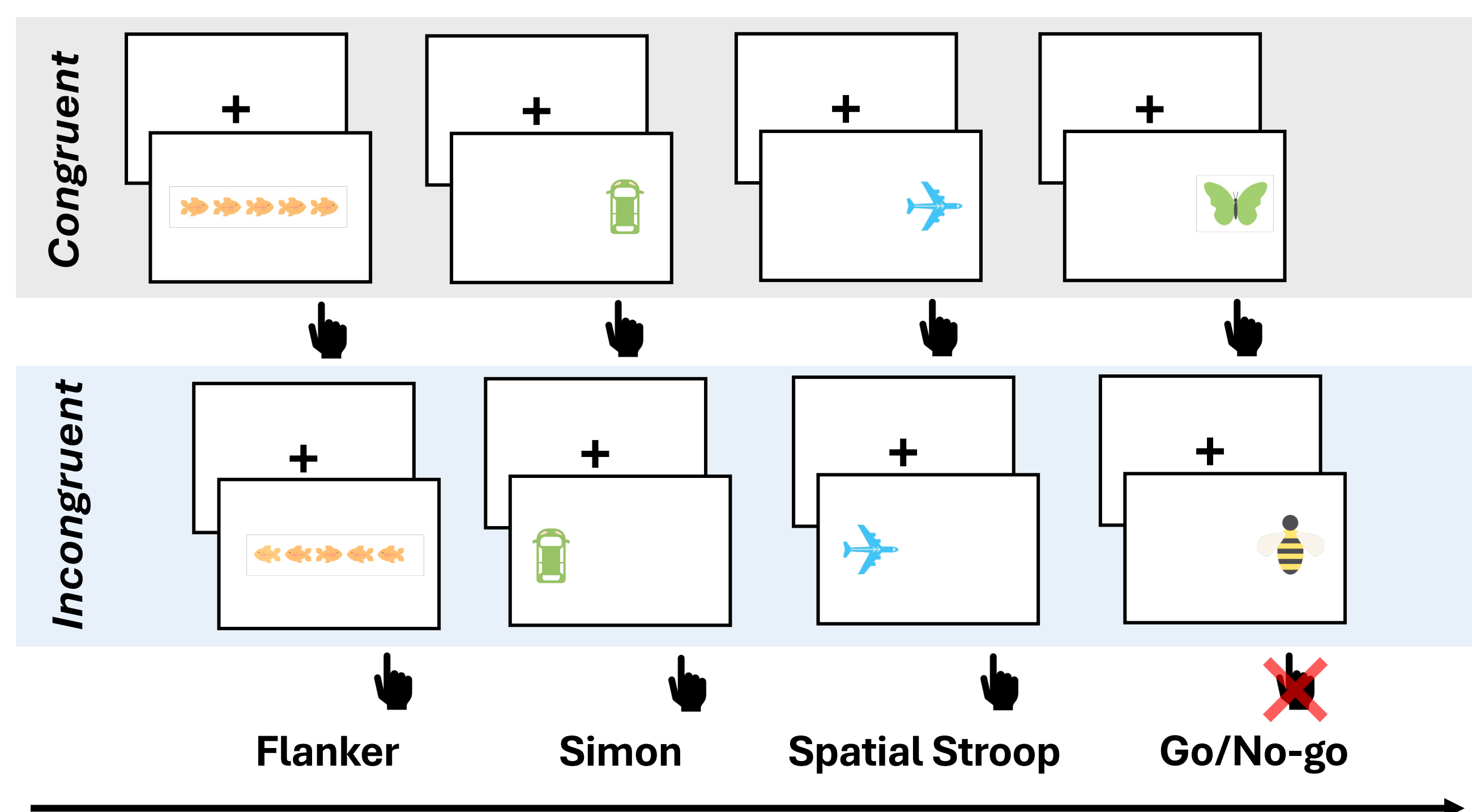


Figure 1: Experiment Design of IC Tasks

Dependent Variables

Indicators: Accuracy (AC), Reaction Time (RT), Inverse Efficiency Score (IES), Drift Rate (DR)
in

Single (Incongruent) and **Difference** (Incongruent – Congruent) score contexts

Results

Q1: What is the best measure to index IC?

Index 1: High Reliability (> 0.7)

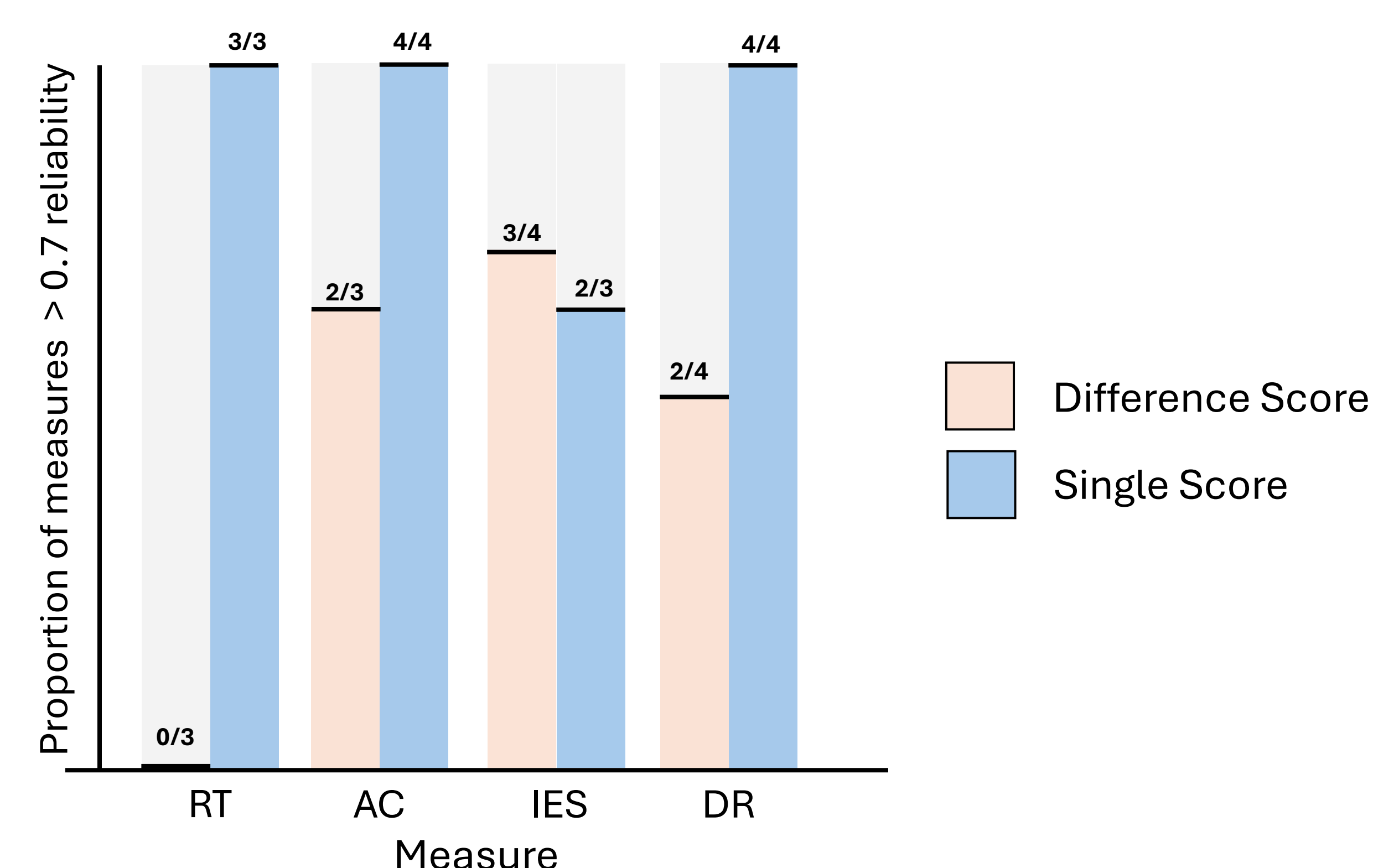


Figure 2: Proportion of measures passing the high reliability threshold (> 0.7)

Index 2: Coherent Latent IC Factor

1. Good Model Fit

Indicators	Conditions	Good Fit
DR	Incongruent	✓
	Difference	✗
IES	Incongruent	✗
	Difference	✓
AC	Incongruent	✗
	Difference	✗
RT	Incongruent	✗
	Difference	✗

Table 1: Factor Model Fit Summary for IC Measures

2. Strong Factor Loadings

Model fit indices (threshold for acceptable fit): $p = .64$ ($> .05$); CFI = 1.00 ($> .95$); TLI = 1.00 ($> .90$); RMSEA = .00 ($< .05$); SRMR = 0.01 ($< .08$)

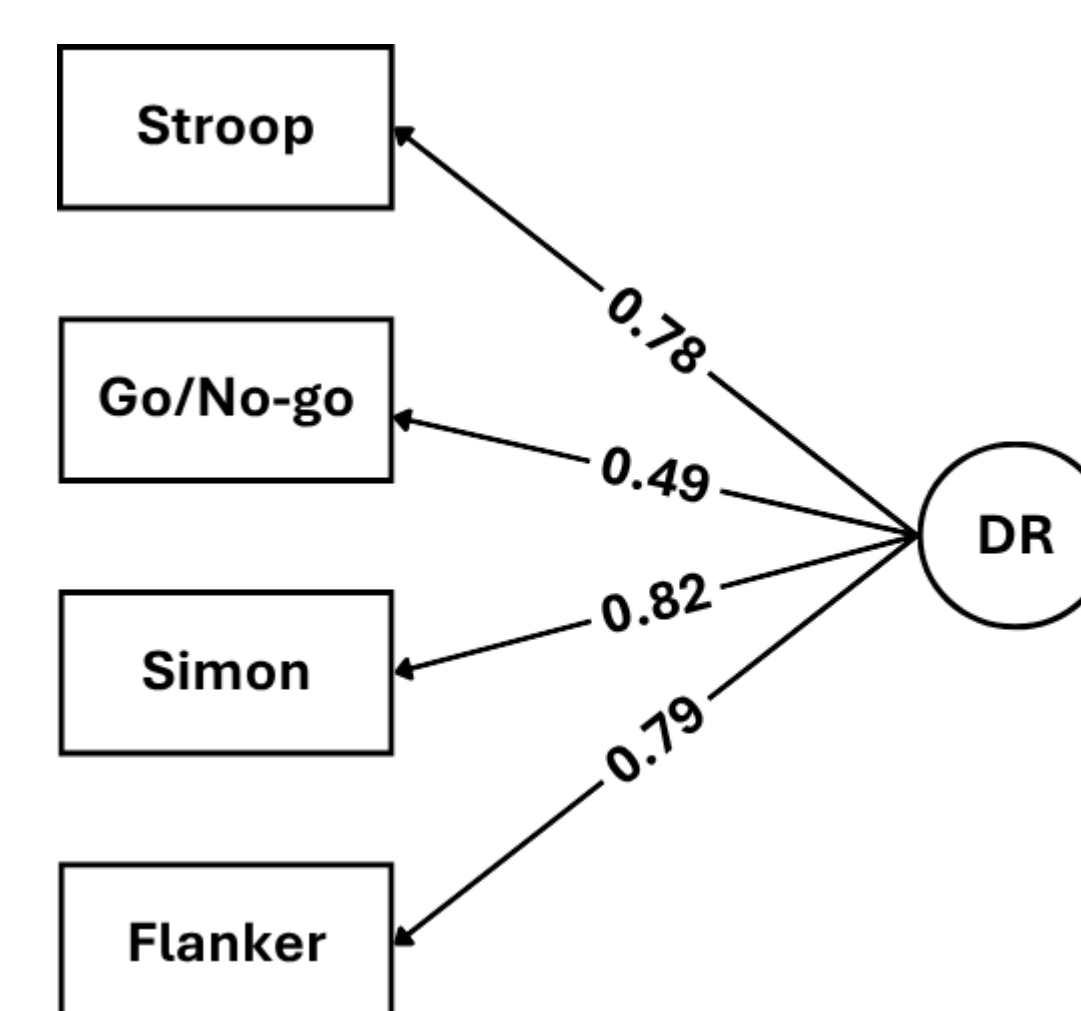


Figure 3: DR single score factor model (all factor loading $p < .05$)

Model fit indices (threshold for acceptable fit): $p = .58$ ($> .05$); CFI = 1.00 ($> .95$); TLI = 1.00 ($> .90$); RMSEA = .00 ($< .05$); SRMR = 0.03 ($< .08$)

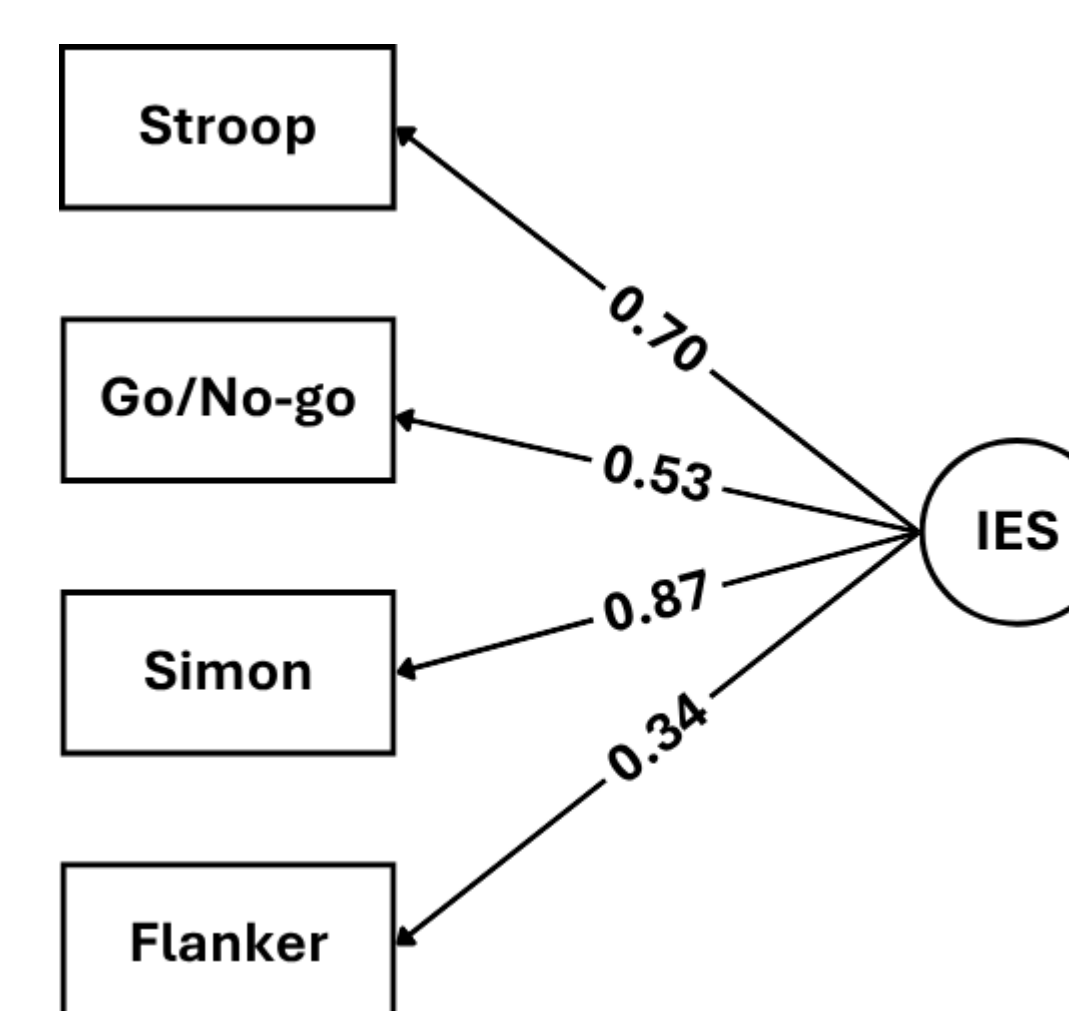
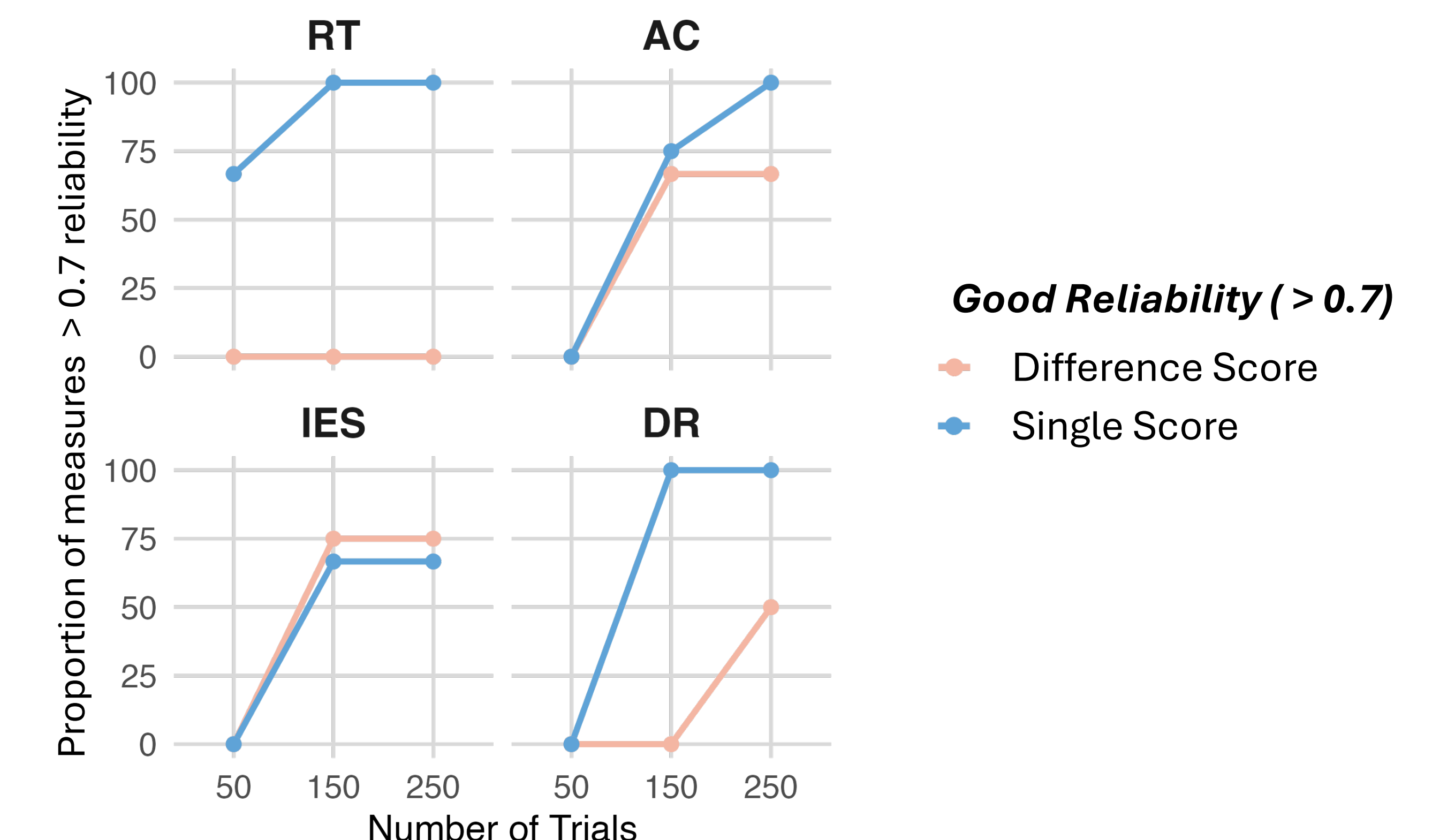


Figure 4: IES difference score factor model (all factor loading $p < .05$)

The most consistently reliably and coherent outcome measure = Incongruent Drift Rate

Results Cont.

Q2: What is the best task design to measure IC?



The task design that resulted in the most reliable outcome measures = 250 trials

Discussion

- Incongruent drift rate showed consistently good reliability across tasks and a strong model fit and factor loadings in the IC factor model.
- Integration-based measures were better able to capture IC abilities.
- Traditional outcome measures (RT & AC), were not consistently reliable or coherent across tasks, therefore researchers should be cautious when using these measures.

Recommendations

When trying to measure IC, consider:

- Using an integration-based outcome measure (DR & IES)
- Using at least 150 trials per IC task

References

1. Diamond, A. (2013). Executive functions. *Annual Review of Psychology*, 64(Volume 64, 2013), 135–168. <https://doi.org/10.1146/annurev-psych-113011-143750>
2. Unsworth, N., Miller, A. L., & Strayer, D. L. (2024). Individual differences in attention control: A meta-analysis and re-analysis of latent variable studies. *Psychonomic Bulletin & Review*. <https://doi.org/10.3758/s13423-024-02516-1>
3. Paap, K. R., Anders-Jefferson, R., Zimiga, B., Mason, L., & Mikulinsky, R. (2020). Interference scores have inadequate concurrent and convergent validity: Should we stop using the flanker, Simon, and spatial Stroop tasks? *Cognitive Research: Principles and Implications*, 5(1), 7. <https://doi.org/10.1186/s41235-020-0207-y>
4. Baggetta, P., & Alexander, P. A. (2016). Conceptualization and operationalization of executive function. *Mind Brain and Education*, 10(1), 10–33. <https://doi.org/10.1111/mbe.12100>
5. Yangüez, M., Bediou, B., Chanal, J., & Bavelier, D. (2024). In search of better practice in executive functions assessment: Methodological issues and potential solutions. *Psychological Review*, 131(2), 402–430. <https://doi.org/10.1037/rev0000434>
6. Rey-Mermet, A., Gade, M., & Oberauer, K. (2018). Should we stop thinking about inhibition? Searching for individual and age differences in inhibition ability. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 44(4), 501–526. <https://doi.org/10.1037/xlm0000450>